
DSC 291 Trustworthy ML

Course Proposal Template (Research Reproduction)

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Project format: Format 1 – Tutorial on Research Reproduction.

Paper: Hernandez et al., *Natural Language Descriptions of Deep Visual Features* (ICLR 2022).

Code / data: <https://github.com/evandez/neuron-descriptions>, <http://milan.csail.mit.edu/>

1 Introduction

1.1 Intro & Motivation

Deep vision models attain state-of-the-art accuracy, yet the visual features encoded by neurons are often opaque. This is a central concern in trustworthy ML, because a model can achieve high accuracy by relying on spurious or unintended features — backgrounds, watermarks, or objects — and such reliance only becomes visible when the model is probed at the level of individual neurons. Existing automated neuron-labeling methods often require a fixed, predefined set of candidate concepts, so they label only a subset of neurons and cannot surface novel feature behaviors. In this paper, we propose MILAN (Mutual-Information-guided Linguistic Annotation of Neurons), which generates open-ended natural-language descriptions of individual neurons in the visual domain (e.g., “grass around animals,” “words and letters on objects”).

This topic sits at the intersection of two pillars of this course, *interpretability* and *robustness*: we use MILAN to analyze the role of different neuron classes in natural language, then use those descriptions to identify and remove neurons sensitive to text features spuriously correlated with class labels. By doing this, we’re able to improve robustness in an image classifier through rich description of automated interpretation procedure.

1.2 Goals and Contributions

MILAN represents each neuron f_i via an exemplar set E_i of input regions on which f_i is highly active, then selects the description that maximizes the pointwise mutual information with E_i :

$$\text{MILAN}(f_i) := \arg \max_d \text{pmi}(d; E_i) = \arg \max_d [\log p(d | E_i) - \log p(d)].$$

The PMI objective forces the description to be both *accurate* for the neuron’s exemplars (high $p(d | E_i)$) and *specific*—not applicable to generic content (low $p(d)$). The two distributions are learned from MILANNOTATIONS, a fine-grained human-annotation corpus released by the authors. The paper’s contributions are: (i) MILAN is the first approach for generating compositional natural language descriptions for interpretability at the level of individual features; (ii) MILAN can do so independently of the model, making it modular, portable; and (iii) three case studies showing that MILAN can support feature-importance analysis, auditing of face-blurred classifiers, and editing of classifiers to remove spurious-text neurons.

2 Proposal

2.1 Reproduction (Experiments, Sections, Tables, Figures)

Among the three case studies in the paper (analysis in Section 5, auditing in Section 6, editing in Section 7), we focus reproduction on **Section 7, *Editing Spurious Features***, for two reasons:

- It simultaneously exercises three central topics from this course—spurious correlations, adversarial robustness, and model editing—and is the case study that most directly demonstrates the value of automated interpretability for trustworthy ML.
- Its experimental loop is intuitive and self-contained: a ResNet18 trained on images with class-name text in the corner learns a text shortcut; MILAN labels each neuron in natural language; neurons whose descriptions contain “text,” “word,” or “letter” are ablated; and the resulting gain on an adversarial test set quantifies the benefit.

Implementation-wise, we will recreate the synthetic spurious dataset, train a ResNet18 from scratch following the hyperparameters in Appendix E, and run MILAN on every residual neuron and the first convolutional layer (1,024 neurons total) using the paper’s released code and pretrained MILAN model (keeping the project tractable for a small team). We will then identify text-selective neurons (those whose MILAN description contains text, word, or letter), rank them by the validation-accuracy drop induced by ablating each one independently on non-adversarial data, and incrementally ablate them while measuring adversarial accuracy.

We will reproduce **Figures 7 and 8**, and report the clean test accuracy, adversarial accuracy before ablation, the number of MILAN-identified text neurons, and the accuracy gain after ablation. As a comparison baseline we will also reproduce the *sort-all* curve in Figure 8, in which all neurons are ranked by validation-accuracy drop and ablated in order.

New experiments There are several experiments that can be made to explore the capability or limitation of MILAN (subject to changes given the time and resources):

- Test generalization across extended type of architecture, e.g. VGGNet and Inception, and test its BERTScore (Extension of Section 4). By doing this, we’re able to validate the generalizability and transferability of MILAN, and proved a complete picture of its effectiveness.
- Applying MILAN to CLIP (perhaps the vision encoder part) to observe the ablation study presented in Section 7. The effect is noteworthy since CLIP has multimodal text-image neurons, investigating via MILAN may shed lights on how CLIP aligns image with text.
- Rerun the BERTScore evaluation or qualitative visual comparison (i.e. Figure 2) broken down by layer depth of model architecture, which captures individual neuron’s behavior on image as model depth or complexity varies.